VISHWANATH GAITONDE PhD

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PROFILE

Accomplished and enthusiastic chemist with 9 years of experience in making and breaking bonds with an unyielding passion for chemistry working in multiple areas of research at academic and industrial setting. Proven ability to manage projects independently while maintaining balance working on team projects in deadline-driven situations. Established technical writing ability demonstrated by publications in journal articles, a book chapter and a patent application.

EDUCATION

- Doctor of Philosophy in Chemistry, *University of Toledo*, *Toledo*, *Ohio*, *USA* **2009-2015**
- Masters of Science in Organic Chemistry, *University of Mumbai, Mumbai*, *India* **2005-2007**
- Bachelors of Science in Chemistry and Biochemistry, *University of Mumbai*, *Mumbai*, *India* **2002-2005**

EXPERIENCE

INDUSTRY

Co-op, *Antiviral Chemistry*, *GlaxoSmithKline*, *NC*, *USA* **2014-Present**

- Molecular modeling with a crystal structure to assist SAR development to improve potency and DMPK properties
- Generated idea from existing SAR to design and synthesize molecules with improved solubility and permeability
- Worked as part of a medicinal chemists team optimizing lead compound to target viral respiratory diseases
- Engaged in brainstorming sessions and performed large-scale synthesis of key intermediates for team use

Chemist, *Duphar Interfran Ltd.*, *Research Support International Ltd.*, *CRO*, *Mumbai*, *India* **2007-2008**

- Successfully generated a library of small molecule pyrazoline-based derivatives for SAR studies
- Synthesized numerous heterocyclic, bicyclic, and other organic intermediates for use in drug discovery
- Independently managed personal projects balancing objectives of the team to achieve target delivery deadlines

Intern, *CSIR-National Chemical Laboratory*, *Pune*, *India* **May - July 2006**

- Synthesized intermediates of (*S*,*R*)-BIPHEMPHOS ligand for use in asymmetric hydroformylation catalysis

ACADEMIC

PhD Candidate, *Dr. Steven J. Sucheck's lab*, *University of Toledo*, *Toledo*, *OH*, *USA* **2010-2014**

Project – 1: Carbohydrate-based methodology

- Synthesized a series of (glucose, galactose, mannose, arabinose, and maltose) *β*-glycosyl amide, key motifs in glycosyl amino acid, employing dinitrobenzenesulfonamide chemistry. Moreover, studied a novel chemoselective amide ligation between glycosyl sulfonamides and thioacid via stereochemical analysis. Successfully developed a synthetic methodology with new route to access *β*-glycosyl amide found in *N*-glycopeptides and several natural products.

Project – 2: Tuberculosis related glycobiology: Investigation of glucan binding site of *Mycobacterium tuberculosis* GlgE (Mtb GlgE)

- Generated maltosyl donor and acceptor building blocks via a concise synthetic route to construct oligosaccharide units using pre-activated glycosylation technique. Further, strategically assembled oligosaccharide units to serve as template to access larger oligosaccharides. These oligosaccharide units are designed as biochemical tools to mimic the *α*-glucan fragments in Mtb GlgE and are utilized to investigate the Mtb GlgE glucan binding site.

Project – 3: Tuberculosis related glycobiology: Identification of active site residues of Mtb GlgE

- Designed and developed synthetic method to access glycosylfluoride compounds as i) potential mechanism-based inhibitors, and ii) to identify the active site residues of Mtb GlgE. Installed fluorine atom at the C-5 position of saccharide which is difficult to access and thereby optimized the photochemically initiated halogenation reaction.

Project – 4: Sustainable chemistry: Bio-based novel polyester development

- Developed a rapid synthetic route producing bis-furan diol (BFD) as a platform molecule to generate novel amorphous polyester. Furthermore, examined BFD for packing, H-bonding and potential estrogenicity via X-ray crystallography. Project culminated in filing a patent "*Amorphous polyester from bio-based bis-furan assembly*"

AWARDS

- Scientist Student Co-op Fellowship, Antiviral DPU – Medicinal Chemistry, GlaxoSmithKline **2014**
- Student Travel Grant, American Chemical Society National Meeting, Toledo Section, ACS **2014**
- Outstanding Teaching Assistant, Department of Chemistry, University of Toledo **2011**

SKILLS

- Medicinal chemistry: interpretation of SAR and crystal structure, designing molecules to improve potency (enzyme, cellular), selectivity, solubility, permeability, and DMPK properties
- Synthetic chemistry: design and synthesis of small molecules, peptides, organic compounds via multi-step synthetic strategies and large scale synthesis
- Carbohydrate chemistry: expertise in glycosylation, orthogonal protection, and range of carbohydrate synthesis
- Knowledge in polymer synthesis and analysis via TGA, DSC, DLS, FTIR, and MALDI-TOF experiments
- Proficient in silica gel flash chromatography, HPLC, crystallization, and distillation techniques
- Competent in ^{1}H, ^{13}C, ^{31}P, ^{19}F, COSY, DEPT, 2D NMR spectra, FTIR and LC-MS analysis
- Trained in eLNB, Spotfire, Helium, MOE, Excel, ChemDraw, and SciFinder software
- Strong background in literature database search for technical writing
- Experienced in GLP, GMP, and GDocP regulatory environment

PRESENTATIONS

1. Gaitonde, V.; Lee, K.; Kirschbaum, K.; Sucheck, S. J. Bio-based bisfuran polymer: Crystal structure of a bisfuran diol (BFD) and synthesis of a low molecular weight amorphous polyester. Presented at the 247th National Meeting of the American Chemical Society, Dallas, TX, March 16-20, 2014; POLY-574.
2. Thanna, S.; Gaitonde, V.; Lindenberger, J. J.; Ronning, D. R.; Sucheck, S. J. Synthesis and study of covalent inhibitors of *Mycobacterium tuberculosis* GlgE. *Abstract of Papers*, 247th National Meeting of the American Chemical Society, Dallas, TX, Mar 16-20, 2014; CARB-80.
3. Gaitonde, V.; Lee, K.; Kirschbaum, K.; Sucheck, S. J. Bio-based bisfuran assembly: Synthesis and crystal structure evaluation with bisphenol A. Presented at the 9th Midwest Carbohydrate and Glycobiology Symposium, University of Toledo, Toledo, OH, Oct 11-12, 2013.
4. Gaitonde, V.; Lee, K.; Kirschbaum, K.; Sucheck, S. J. Synthesis of 5,5'-(propane-2,2-diyl)bis(furan-2,5-diyl)dimethanol from biomass. *Abstract of Papers*, 244th National Meeting of the American Chemical Society, Philadelphia, PA, Aug 19-23, 2012; CARB-22.
5. Gaitonde, V.; Sucheck, S. Synthesis of *β*-gycosyl amides via dNBS-thioacid ligation. Presented at the 7th Midwest Carbohydrate and Glycobiology Symposium, Michigan State University, East Lansing, MI, Sept 16-17, 2011.
6. Gaitonde, V.; Sarkar, S.; Lodzinski, A. J.; Sucheck, S. J. Synthesis of *N-β*-glycosyl dinitrobenzenesulfonamides and their reactions with thioacids for the formation of *β*-glycosyl amides. Presented at the 6th Midwest Carbohydrate and Glycobiology Symposium, University of Toledo, Toledo, OH, Sept 24-25, 2010.

PUBLICATIONS

1. Catalano, J.; Gaitonde, V.; Leivers, A. A phenoxide SNAr leaving group strategy for the facile preparation of 7-amino-3-arylpyrazolo[1,5-a]pyrimidines (manuscript in preparation) **2015**
2. Bouhall, S.; Gaitonde, V.; Sucheck, S. J. Synthesis of oligosaccharide mimics of the *α*-glucan fragment and its interaction with the glucan binding site of *Mycobacterium tuberculosis* GlgE (manuscript in preparation) **2015**
3. Thanna, S.; Lindenberger, J.; Gaitonde, V.; Ronning, D.; Sucheck, S. J. Synthesis of 2-deoxy-2,2-difluoro- *α*-maltosyl fluoride and its X-ray structure in complex with *Streptomyces coelicolor* GlgEI-V279S (submitted) **2015**
4. Gaitonde, V.; Sucheck, S. J. Antitubercular drug based on carbohydrate derivatives. In *Carbohydrate Chemistry*: *State-of-the-art and challenges for drug development*; Cipolla, L., Ed.; Imperial College Press: London (accepted) **2015**
5. Gaitonde, V.; Sucheck, S. J. Amorphous polyester from bio-based bis-furan assembly. U.S. Patent PCT 14/621,520 filed on February 13, **2015**
6. Gaitonde, V.; Lee, K.; Kirschbaum, K.; Sucheck, S. J. Bio-based bisfuran: synthesis, crystal structure and low molecular weight amorphous polyester. *Tetrahedron Lett*. **2014**, *55*, 4141-4145
7. Gaitonde, V.; Sucheck, S. J. Synthesis of *β*-Glycosyl amides from *N*-Glycosyl dinitrobenzenesulfonamides. *J. Carbohydr. Chem*. **2012**, *31*, 353-370 (*Special issue*)